

An overview of arable land use for the world economy: From source to sink via the global supply chain

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ABSTRACT

As an extension of a previous work (Chen and Han, 2015a), this study explored the arable land use of the world economy from source of exploitation to sink of final consumption via the global supply chain, by means of embodiment accounting that includes the indirect feedbacks associated with both intermediate and primary inputs. In magnitude, the global transfer of arable land use is estimated to be around 40% of the total direct exploitation. The connections as well as imbalances of major economies in intermediate and final trades of arable land use are discussed. Canada, Australia, Argentina, Pakistan and African regions turn out to have a massive deficit of arable land use in both intermediate and final trades. In contrast, the United States, Japan, Mainland China, the United Kingdom, Germany and France obtain a surplus of arable land use in both intermediate and final trades by land displacement in those net exporters. Indices in terms of arable land use self-sufficiency rate by source and that by sink are devised. For India as the biggest source region, around 20% of the arable land resources exploited locally are for final consumption abroad. For the United States as the largest sink region, around 40% of its arable land use originates from foreign regions led by Canada. For Japan as the biggest net importer in both intermediate and final trades, over 90% of its arable land use comes from foreign economies led by African and Asian regions. For sustained development, regions are suggested to be more adapted to the global supply chain based on their behaviors in both intermediate and final trades of arable land use.

1. Introduction

Arable land, defined as land suitable for growing crops (OxfordDict, 2018), remains the precondition for delivery of food that feeds all humans. Arable land has been entitled as “mother” in some eastern countries such as China and India that were rooted on agricultural foundation since ancient times (Long, 2014). Similarly, in some western country like France, physiocrats such as Francois Quesnay and Anne Turgot have attached supreme importance to arable land by regarding agriculture as the sole source of the wealth of nations (Quesnay, 1758; Turgot, 1793).

Nowadays arable land use has been greatly threatened. The world is changed along with the overwhelming wave of urbanization and booming population. Much that once was is gone. According to Cameron et al. (2015), the world's arable land area has encountered a drastic decline by one-third during the last four decades, due to soil erosion, chemical and physical deterioration, desertion and encroaching human inhabitation (Hubacek and Sun, 2001; Lasanta et al.,

2017; Weinzettel et al., 2013). Currently, a number of policies have been implemented worldwide for purposes of sustainable use of arable land resources, such as the red line set in the *Eleventh Five-Year Plan of China* that regulates the minimum (180 million hectares exactly) of domestic total arable land area (StateCouncil, 2006), the *Common Agricultural Policy* in the European Union (Grant, 1997) and *Soil and Water Resources Conservation Act* (USC, 1977) in the United States.

Nevertheless, a local perspective is mostly adopted in these policy packages to view the arable land use of a region. In fact, within the globalized world, interregional trade not only transfers good or services from one region to another, but could become an effective means for a region to invisibly acquire arable land use from other areas. As the sliced-up global supply chain leads to the separation of production and consumption (Dunford, 2017; Dunford et al., 2016; Meng et al., 2016), one region may enjoy the welfare denoted by land resources via importing massive land-intensive goods or services from regions outside its territorial border, which is sometimes referred to as “land grab” (Cecilie and Anette, 2010; Gorgen et al., 2009). For those developed

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economies with affluent lifestyles, the consumption-oriented development mode is associated with a heavy dependency on both domestically-manufactured as well as foreign goods or services to fulfill the residential demands. As a result, a burden is placed on both their domestic and foreign arable land resources (Verhoeve et al., 2015). For instance, according to Yu et al. (2013), the land use hidden in the imported commodities of Japan is in magnitude several times as much as Japan's direct land exploitation.

With the virtual shrinking of spatial distances and accelerating instant communication, this interregional transfer of arable land use will become more frequent since all world regions are being integrated into a united organism (Han and Chen, 2018). The production model of iPhone may be an illuminating example, which makes full use of the global supply chain to integrate the resources and technology all over the world. Currently, iPhone has several hundred suppliers for its elements all over the world (Apple, 2018; Chen et al., 2017; Minasians, 2017): R&D department in the United States is in charge of the product design; manufacturers in Japan are in charge of the production of major parts; corporations in Korea supply electronic chips; producers in Singapore, Malaysia, Thailand and Philippines provide back-end components; Foxconn factories in Mainland China assemble iPhone products and ship it to the United States for online sales to the world consumers. Therefore, in order to have a deep understanding of the arable land use of a region, it is necessary to put all world regions in the global context to explore the panorama of the trade connections between them in terms of arable land use transfer.

The academic history of virtual land transfer in trade may be traced back to 1965 when Borgstrom (1965) raised the notion of “Ghost Acreage” to describe the cropland use induced to produce the traded agricultural products. Later in 1996, the concept of ecological footprint was introduced by Wackernagel and Rees (1996) to analyze the bio-productive area of land that is directly and indirectly required to sustain the resource consumption of a region, which has been widely applied in subsequent studies (Erb, 2004; McDonald and Patterson, 2004; Wackernagel et al., 1999; van den Bergh and Verbruggen, 1999). As pointed out by some scholars, the ecological footprint is a highly aggregated indicator by putting the different kinds of resources into a unified measurement (Qiang et al., 2013). Besides, the interdependency between the economic activities finds no reflection in the ecological footprint accounting (Wiedmann et al., 2006). Hence, other concepts termed as “virtual land” (Würtenberger et al., 2006) and “land displacement” (Meyfroidt and Lambin, 2009) have been proposed to specifically analyze the land use embedded in interregional trade, which have been extensively applied for economies on different scales (Chen and Han, 2015b; Guo and Shen, 2015; Hubacek and Sun, 2001; Meier et al., 2014; Meyfroidt et al., 2010; Verhoeve et al., 2015; Wilting and Vringer, 2009). Especially, by introducing global multi-regional input–output (MRIO) tables that have been in recent years unveiled for the world economy, a number of studies have been carried out to trace the interregional transfer of virtual land through the global supply chain (Weinzettel et al., 2013; Wilting and Vringer, 2009; Yu et al., 2013), which reveal the remarkable amount of land use embedded in trade. According to Weinzettel et al. (2013), the land displacement through international trade was estimated to be strikingly in magnitude up to around a quarter of the global land use of the world economy in 2004.

These global studies have contributed greatly to increasing our knowledge on interpreting the arable land use embedded in the inter-regional trade flows, while they do not distinguish the arable land use embodied in intermediate trade and that in final trade. Statistics by Johnson and Noguera (2011) have highlighted the dominant role played by intermediate trade, which accounts for over two-thirds of the global total trade volume. In view of this, by means of the embodiment accounting supported by multi-regional input–output accounts, Chen and Han (2015a) have for the first time directed equal attention to arable land use embodied in trade of intermediate products (namely goods or services used for production activities) as well as that

embodied in trade of final products (namely goods or services used as final demand). The results reveal that global total trade volume of arable land use is in magnitude equivalent to around one-third of global total arable land use, while intermediate trade volume of arable land use is twice as much as the final trade volume. The products used as final demand are regarded as sinking into the society in the work by Chen and Han (2015a), which does not take into consideration of the external feedback from the society to the economic system. For the world economy, final demand not only includes consumptive activities including household consumption, government consumption, and consumption of non-profit organization serving household, but also covers fixed capital formation, changes in inventories, and valuables acquisition. Those that could be treated as the sink of arable land use are the goods or services used for final consumption (including household consumption, government consumption, and consumption of non-profit institutions serving households), which are supposed to be “consumed” and ultimately leave the economic system (Chen and Chen, 2013a). While for the products used as the rest of final demand (final demand excluding final consumption), they are capital goods that will get back into the economy as primary inputs in support of economic production (Marx, 1867; Bullard and Herendeen, 1975b). Bullard and Herendeen (1975b) have made preliminary attempts to include indirect feedback associated with capital goods when undertaking embodied analysis for the United States economy. Similarly, regarding arable land use accounting of the world economy, apart from the indirect feedback associated with intermediate inputs, due attention shall be paid to arable land use embodied in the capital goods that will enter the economy in the form of primary inputs.

As an extension of a previous work (Chen and Han, 2015a), this study analyzes the interregional transfer of arable land use from source of exploitation to sink of final consumption for the world economy by using typical statistics. By incorporating the indirect feedbacks associated with both primary and intermediate inputs, an embodiment accounting model is developed from the one in the previous study. Arable land use embodied in interregional trade is presented, with intermediate and final trade imbalances being discussed in detail for major economies.

2. Methodology and data source

2.1. Embodiment accounting

By integrating the embodiment theory in systems ecology by Odum (1983, 1995) into the biophysical balance model raised by the energy research group (Bullard, 1975; Bullard and Herendeen, 1975; Hannon et al., 1983; Herendeen, 1973, 1974, 1978, 1981, 2004) in the University of Illinois, Chen and his colleagues have generalized the embodied energy accounting to embodiment accounting, which has become a well-established tool in tracking the ecological endowments such as energy use (Chen and Chen, 2011c; Wu and Chen, 2017b, 2018), land use (Chen and Han, 2015b; Chen et al., 2018; Han and Chen, 2018), water use (Chen and Chen, 2013b), carbon emissions (Chen and Chen, 2011a,b; Chen and Zhang, 2010; Zhang and Chen, 2010) and mercury emissions (Chen et al., 2016; Li et al., 2017), embodied in interregional trade flows within the world economy.

For the world economy as simulated by the global MRIO account, it is regarded to be consist of m world regions with each region comprised of n economic sectors, as presented in Table 1. The embodiment accounting model developed in this study for the world economy and the one in the previous work are both illustrated in Fig. 1. For the embodiment accounting model raised in the previous work, the biophysical balance for an economic sector is that resource use embodied in the sectoral output is equal to the sum of resource use embodied in the goods or services used as intermediate inputs into this sector plus the exogenous resource inputs from the environment, just as illustrated in Fig. 1(a). While the exogenous resource inputs coming from the

Table 1
The structure of the ecological MRIO account.

Output Input			Intermediate use						Final demand							
			Region 1		...	Region m		Region 1				...	Region m			
			Sector 1	...	Sector n	...	Sector 1	Sector n	Household consumption	Non-profit institutions consumption	Government consumption	Rest of final demand (capital goods)	...	Household consumption	Non-profit institutions consumption	Government consumption
Intermediate input	Region 1	Sector 1	z_{ij}^{rs}						f_i^r							
		...														
		Sector n														
	...															
	Region m	Sector 1														
		...														
Sector n																
Primary inputs			p_j^s													
Exogenous environmental inputs			u_j^s													

environment could be regarded as environment's biophysical support for the economic system, resource use embodied in intermediate inputs is the internal feedback within the economic system (Chen and Wu, 2017). Correspondingly, the embodied intensity of a sector's products could be obtained as the resource use directly and indirectly required to produce per monetary unit of the sectoral output (Herendeen, 1973; Bullard, 1975; Wei et al., 2018; Wu and Chen, 2017a). It needs to be noted that this intensity applies to the total output of the sector, namely both intermediate products and final products (Bullard and Herendeen, 1975b; Bullard and Pilati, 1976; Wu et al., 2016, 2018). Under this embodiment accounting model, the process could be comprehended as follows: resource use comes into the economic system through environment via economic sectors; a larger quantity of resource use circulates in the economy; a quantity equivalent to the exogenous resource inputs sinks into the society via final demand (Herendeen, 1973).

As previously mentioned, products used as the rest of final demand are bound to come into the economic system again, as already been noticed in a number of studies (Bullard and Herendeen, 1975a,b). The biophysical balance for a sector in terms of resource use has thus changed in the embodiment accounting model developed in this study, as illustrated in Fig. 1(b). The resource use received by an economic sector is comprised of three parts: exogenous resource inputs provided by the environment; resource use embodied in the intermediate inputs; resource use embodied in primary inputs. The summation of these three parts is equal to the resource use embodied in the sectoral total output. Meanwhile, a sector's output could be categorized into three parts, namely products used as intermediate inputs, those used as final consumption, and those used as the rest of final demand (capital goods). Resource use embodied in intermediate products could be regarded as internal feedback within the world economy while that embodied in primary inputs could be treated as external feedback from the society.

Only resource use embodied in products used as final consumption could be treated as sinking into the society. Under this embodiment accounting model, the picture could be described as: resource use comes into the economic system through environment via the economic sectors; a remarkable quantity of resource use circulates in the economy; a considerable quantity of resource use comes into the economy as external feedback from the society; a quantity that equals the exogenous resource inputs sinks into the society via final consumption. In general, the sole difference from the previous model is that the embodiment accounting model developed in this study directs equal attention to indirect feedback associated with primary inputs and that with intermediate inputs.

2.2. Algorithm

Illustrated in Fig. 2 is the embodied resource flow for Sector j in Region s . The biophysical balance equation in terms of resource use could be thus established as:

$$u_j^s + \varepsilon_p p_j^s + \sum_{r=1}^m \sum_{i=1}^n \varepsilon_i^r z_{ij}^{rs} = \varepsilon_j^s x_j^s, \quad (1)$$

where u_j^s represents the exogenous resource inputs from the environment into Sector j in Region s ; p_j^s is the primary inputs from the society into Sector j in Region s ; ε_p is the embodied intensity of the primary inputs; z_{ij}^{rs} is the intermediate inputs from Sector i in Region r into Sector j in Region s ; ε_i^r is the embodied intensity of products manufactured by Sector i in Region r ; ε_j^s is the embodied intensity of the products generated by Sector j in Region s ; x_j^s is the sectoral output of Sector j in Region s , comprising $\sum_{r=1}^m \sum_{i=1}^n z_{ji}^{sr}$ (the amount of sectoral output of Sector j in Region s that is used as intermediate inputs to all

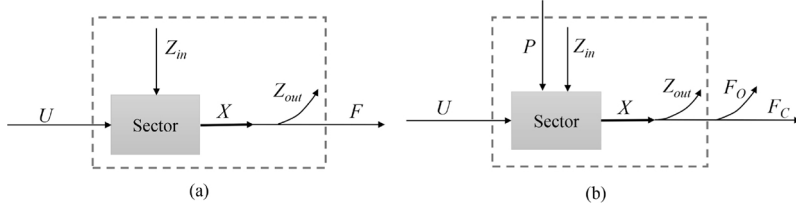


Fig. 1. The embodiment accounting model in (a) the previous work and (b) in this work.

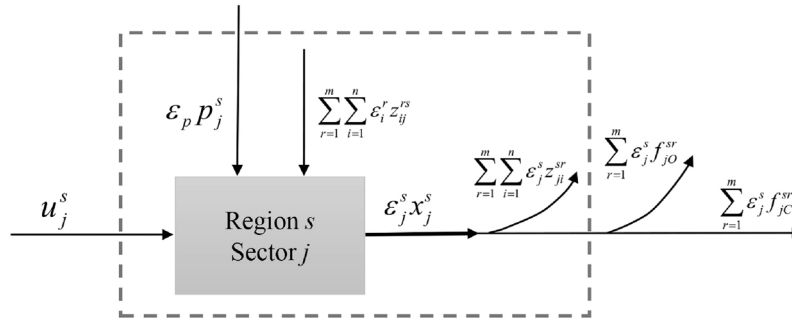


Fig. 2. Resource use flows for Sector j in Region s within the world economy.

economic sectors), $\sum_{r=1}^m f_{jC}^{sr}$ (the amount of sectoral output of Sector j in Region s that is used as final consumption), and $\sum_{r=1}^m f_{jO}^{sr}$ (the amount of sectoral output of Sector j in Region s that is used as the rest of final demand). By transforming Eq. (1) into matrix form could we obtain:

$$U + \varepsilon_p P + \varepsilon Z = \varepsilon \hat{X}, \quad (2)$$

in which $U = [u_j^s]_{1 \times mn}$; $P = [p_j^s]_{1 \times mn}$; $Z = [z_{ij}^{rs}]_{mn \times mn}$; $\varepsilon = [\varepsilon_i^r]_{1 \times mn}$; $\varepsilon_p = [\varepsilon_p]_{1 \times 1}$; $\hat{X}$ is the diagonal matrix for $X (= [x_j^s]_{1 \times mn})$. It is worth noting that ε_p is a scalar, which means that primary inputs into different economic sectors are regarded to have the same embodied intensity. The reason will be explained later.

Meanwhile, for the world economy as represented by the global input-output account, it may be viewed as a flow field. First, there exists a monetary balance between the inputs and outputs of the economic sectors. The total monetary cost of the products used as final demand equal the monetary cost of the primary inputs. Second, the biophysical balance of resource use that corresponds to the global input-output account could also be maintained for the world economy. While exogenous resource inputs from the environment flow into the economic system as the source, the resource use embodied in the products used as final consumption leaves the economic system as the sink. Meanwhile, in the global input-output account, the redistribution matrix provides the means for all the capital goods to get into the economy as primary inputs from the society. Since a normalization of the economy has been made in the static input-output account, the capital goods form into a circulation in the balance account for the world economy. Therefore, resource use embodied in the products used as the rest of final demand (namely capital goods) equals that embodied in the primary inputs as social support into the world economy. Currently, though the socio-economic matrixes for some nations and regions have been established (Hanson and Robinson, 1991; Graham and Round Jeffrey, 2005; Stone, 1973), the one that corresponds to the multi-regional input-output account of the world economy is lacking. As a result, the connection between final demand and primary inputs is unknown to us. As a preliminary treatment, all the primary inputs are treated to be with the same embodied intensity, regardless of the sectoral belonging or type. The balance equation for resource use embodied in all the products used as the rest of final demand and that embodied in the total primary inputs could be constructed as:

$$\varepsilon_p \sum_{s=1}^m \sum_{j=1}^n p_j^s = \sum_{s=1}^m \sum_{j=1}^n \sum_{r=1}^m \varepsilon_j^s f_{jO}^{sr}. \quad (3)$$

By transforming Eq. (3) into matrix form could we get:

$$\varepsilon_p P_{sum} = \varepsilon F_O, \quad (4)$$

where P_{sum} is a scalar value that adds up the monetary cost of all primary inputs; F_O is a column vector that represents the rest of final demand. Combining Eqs. (2) and (4) yields:

$$\varepsilon = U \left(\hat{X} - Z - \frac{1}{P_{sum}} F_O P \right)^{-1}. \quad (5)$$

The resource use of Region s , namely resource use embodied in the products used as final consumption (UEC) in Region s , could be expressed as:

$$UEC^s = \sum_{r=1}^m \sum_{i=1}^n \varepsilon_i^r f_{iC}^{rs}. \quad (6)$$

For comparison, the exogenous resources exploited from the environment (EUE) in Region s are denoted as:

$$EUE^s = \sum_{j=1}^n u_j^s. \quad (7)$$

The resource use associated with trade is classified into resource use embodied in intermediate trade and that embodied in final trade. Resource use embodied in intermediate imports (UEII) and that embodied in final imports (UEFI) of Region s are respectively formulated as:

$$UEII^s = \sum_{r=1(r \neq s)}^m \sum_{i=1}^n \sum_{j=1}^n \varepsilon_i^r z_{ij}^{rs}, \quad (8)$$

and

$$UEFI^s = \sum_{r=1(r \neq s)}^m \sum_{i=1}^n \varepsilon_i^r f_i^{rs}. \quad (9)$$

Meanwhile, resource use embodied in intermediate exports (UEIE) and that embodied in final exports (UEFE) of Region s are respectively formulated as:

$$UEIE^s = \sum_{r=1(r \neq s)}^m \sum_{j=1}^n \sum_{i=1}^n \varepsilon_j^s z_{ji}^{sr}, \quad (10)$$

and

$$UEFE^s = \sum_{r=1(r \neq s)}^m \sum_{j=1}^n \varepsilon_j^s f_j^{sr}. \quad (11)$$

Therefore, resource use embodied in total imports of Region s and that embodied in total exports of Region s are respectively expressed as:

$$UEI^s = UEII^s + UEFI^s = \sum_{r=1(r \neq s)}^m \sum_{i=1}^n \left[\sum_{j=1}^n (\varepsilon_i^r z_{ij}^{rs}) + \varepsilon_i^r f_i^{rs} \right], \quad (12)$$

and

$$UEE^s = UEIE^s + UEFE^s = \sum_{r=1(r \neq s)}^m \sum_{j=1}^n \left[\sum_{i=1}^n (\varepsilon_j^s z_{ji}^{sr}) + \varepsilon_j^s f_j^{sr} \right], \quad (13)$$

Resource use embodied in intermediate trade balance (UEITB) represented by net intermediate imports of resource use, and that embodied in final trade balance (UEFTB) represented by net final imports of resource use are respectively expressed as

$$UEITB^s = UEII^s - UEIE^s, \quad (14)$$

and

$$\text{UEFTB}^s = \text{UEFF}^s - \text{UEFE}^s. \quad (15)$$

Meanwhile, it is worth noting that there exists the following equation between resource use embodied in final consumption as sink and exogenous resources exploited as source, expressed as:

$$\sum_{s=1}^m \text{UEC}^s = \sum_{s=1}^m \text{EUE}^s. \quad (16)$$

2.3. Data sources

By combining the recent statistics, the 2012 world economy is chosen as the case for this study. The corresponding global MRIO account is adopted from Eora database (Lenzen et al., 2013, 2012), which models the world economy as a 189-region, 26-sector system. The names of the associated regions and sectors are respectively presented in Appendices A and B. Meanwhile, arable land use data for world economy 2012 come from the database unveiled by Food and Agricultural Organization of the United States (FAO, 2017), which presents a time-series data for arable land use of different regions all over the world. Arable land use of a region is totally allocated to the agricultural sector given that the exploitation of arable land mainly happens in this sector. Population data are adopted from the WorldBank (2017).

3. Results

3.1. Arable land use of each region within the world economy

The arable land use embodied in global final consumption equals the arable land exploitation of the world economy, which amounts to 1406.60 million hectares (ha). The indirect feedback associated with global primary inputs, namely arable land use embodied in primary inputs of the world economy, reaches an amount of 199.44 million ha, which is in magnitude around 14.18% of the arable land use embodied in global final consumption. The arable land use of each region covered within the Eora global MRIO table is illustrated in Fig. 3, together with the arable land exploitation of the regions. The associated numerical values are attached in Appendix C. The arable land use of the United States, denoted by arable land use embodied in the goods and services used for its final consumption, is the largest among all regions covered within global input–output account, reaching an amount of 200.62 ha. India ranks the second place with 139.69 million ha of arable land use, followed by Russia (122.65 million ha), Mainland China (104.29 million ha), Brazil (69.35 million ha), Japan (49.12 million ha) and Germany (34.68 million ha). With regard to the arable land exploitation, India occupies the first place with 156.55 million ha of arable land exploited, followed closely by the United States (155.11 million ha), Russia (119.75 million ha), Mainland China (105.92 million ha), Brazil (72.61 million ha), Australia (47.11 million ha), Argentina (39.75

million ha) and Nigeria (35.00 million ha), etc. The sectoral contributions to the arable land use of the five leading users are presented in Appendix D.

Fig. 4 captures the per-capita arable land use of each region. As witnessed, the per-capita arable land use of the United States, Canada, Australia, Japan, Germany and France are respectively 0.64 ha, 0.81 ha, 0.97 ha, 0.39 ha, 0.43 ha and 0.44 ha, which are several times larger than the world average (0.20 ha/capita). For the developing economies including Mainland China and India, the per-capita arable land use are 0.08 ha and 0.11 ha, which are respectively only around 40% and half of the world average. The divergence between the per-capita arable land use of the developing regions and that of the developed regions implies the big gap of their residential welfare from a biophysical perspective.

3.2. Arable land use embodied in trade

Fig. 5 depicts the arable land use embodied in trade. The global total trade volume of arable land use, namely the sum total of arable land use embodied in the imports of each region, is calculated to be 570.44 million ha, in magnitude up to around 40% of the global total exploitation of arable land. Meanwhile, the global total intermediate trade volume of arable land use amounts to 381.18 million ha, over twice the global final trade volume, revealing the importance of intermediate trade in interregional transfer of arable land use.

As witnessed, the largest ten importers of arable land use are the United States with a total import volume of 63.37 million ha, Germany (46.85 million ha), Mainland China (45.72 million ha), Japan (39.67 million ha), Netherlands (23.90 million ha), France (23.37 million ha), the United Kingdom (22.56 million ha), Italy (20.56 million ha), Russia (19.93 million ha) and India (16.40 million ha). As for arable land use embodied in exports, the ten leading exporters are the United States exporting 46.71 million ha of arable land use, Canada (32.16 million ha), Germany (30.23 million ha), Australia (25.27 million ha), Mainland China (24.44 million ha), India (23.93 million ha), Netherlands (19.23 million ha), Argentina (17.97 million ha), France (15.69 million ha) and Ethiopia (15.39 million ha). As could be seen, regions such as the United States, Germany, Mainland China and Netherlands serve as hubs in interregional trade, which act as leading roles in both global importing and exporting market.

Fig. 6 illustrates arable land use embodied in trade balance of each region. Japan, Mainland China, the United States, Germany and the United Kingdom turn out to be the biggest five net importers, whose net imported arable land use amount to 33.73 million ha, 21.29 million ha, 16.67 million ha, 16.62 million ha and 14.84 million ha respectively. Australia, Canada, Argentina, Ethiopia, Pakistan, Tanzania, Nigeria, India, Myanmar and Belarus are revealed to the top ten net exporters, whose net exported arable land use respectively reach an amount of 20.43 million ha, 18.41 million ha, 15.81 million ha, 15.23 million ha,

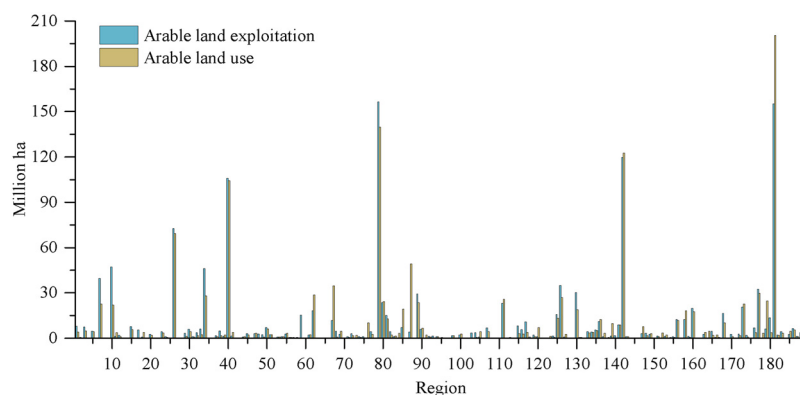


Fig. 3. Arable land use and direct arable land exploitation of each region.

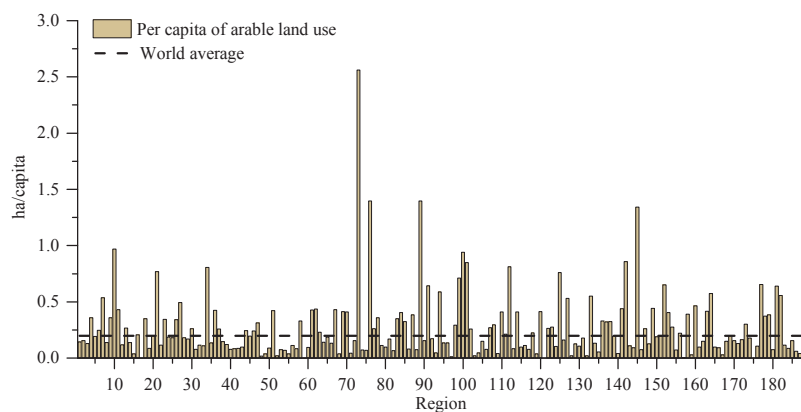


Fig. 4. Per-capita arable land use of each region.

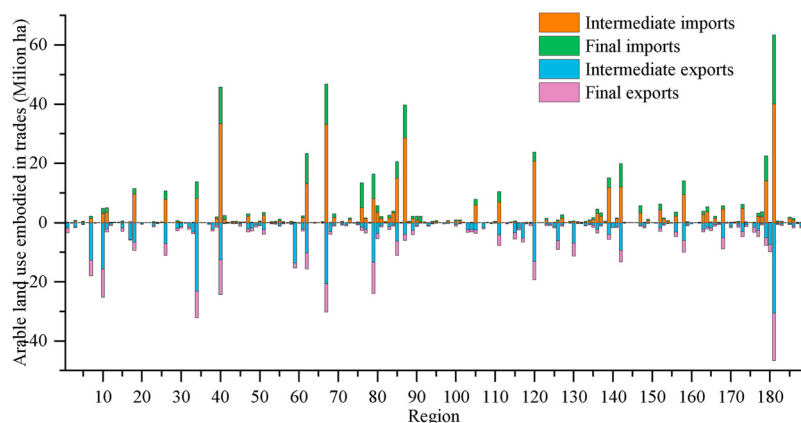


Fig. 5. Arable land use embodied in trades.

10.64 million ha, 9.78 million ha, 8.17 million ha, 7.53 million ha, 6.58 million ha and 5.51 million ha.

Sectoral contributions to intermediate trade of the top five net intermediate importers and exporters of arable land use are illustrated in Fig. 7. Appendix B gives the details of sector aggregation. Japan ranks in the first place in net intermediate imports of arable land use, whose intermediate imports are several times as many as its intermediate exports. At witnessed, Japan imports lots of agricultural products and food & beverage products for domestic producing activities. Within the 28.57 million ha of Japan's intermediate imports of arable land use, imported agricultural products account for 64.97% of the total,

followed by the imported food & beverage products (16.39%), etc. Similarly, for Mainland China, Germany, the United States, and the United Kingdom, foreign agricultural industry also takes dominant shares in their intermediate imports of arable land use, reaching 54.72%, 51.72%, 43.70% and 71.28% respectively. With regard to the intermediate exports, while agricultural sector holds accountable for a meager portion of the intermediate exports of arable land use for Japan, Mainland China as well as Germany, it contributes to over half of those for the United States. Meanwhile, for Tanzania, Argentina, Australia, Ethiopia and Canada, they are characterized with an intensive imbalance of arable land use in intermediate trade, whose intermediate

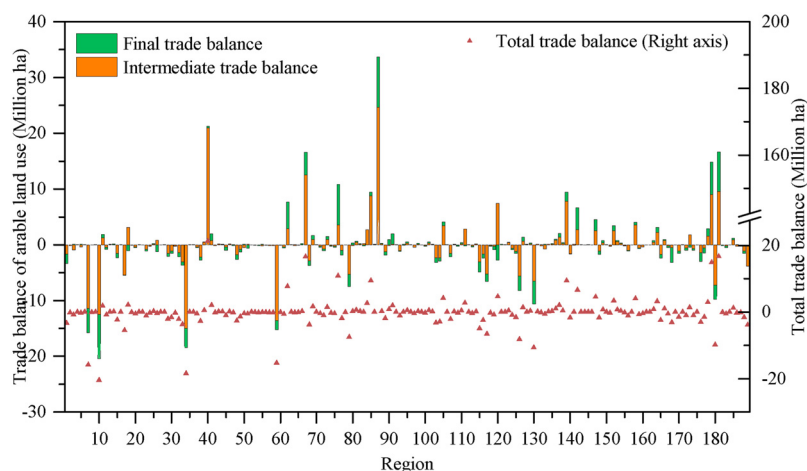


Fig. 6. Arable land use embodied in trade balance.

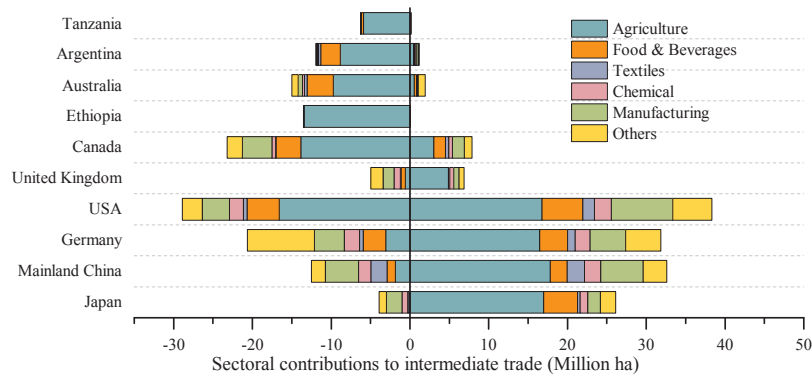


Fig. 7. Sectoral contributions to intermediate trade of the top five net intermediate importers and exporters of arable land use.

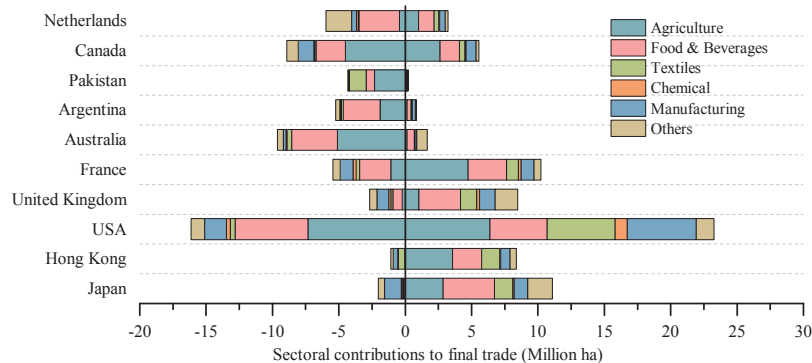


Fig. 8. The sectoral contributions to final trade of the top five net final importers and exporters of arable land use.

exports are nearly one or several orders of magnitude larger than their intermediate imports. Agricultural sector is responsible for 93.97%, 73.95%, 64.91%, 99.87% and 59.72% of their intermediate exports of arable land use, respectively.

Fig. 8 presents the sectoral contributions to final trade of the top five net final importers and exporters of arable land use. While Japan, Hong Kong, the United States, the United Kingdom and France are the five leading net importers in final trade of arable land use, Australia, Argentina, Pakistan, Canada and Netherlands are illustrated as the five leading net exporters. For Japan, foreign food & beverage sector shares the largest proportion (34.91%) of the arable land use embodied in Japan's final imports, followed by agricultural sector (25.64%), etc. A similar situation is also witnessed for the United Kingdom, whose imported food & beverage products account for 37.09% of the arable land use embodied in its final imports. While for the United States, Hong Kong and France, foreign agricultural industry remains the biggest contributor to their final imports of arable land use. For Netherlands, Canada and Australia as notable net exporters of arable land use in final trade, arable land use embodied in the final exports are respectively around twice, one and a half times, and six times that embodied in the final imports. Meanwhile, the final exports of arable land use for Australia as well as those for Canada are mainly induced by the massive exported agricultural products for foreign final consumption while that for Netherlands are mainly due to the outflow of food & beverage products.

3.3. Intertwined trade relations between major economies

In this section, the intertwined trade relation of arable land use between the major economies are presented, by splitting the world economy into fourteen regions including China (containing Mainland China, Hong Kong, Macao, and Taiwan), ASEAN (Association of Southeast Asian Nations), Japan, other Asia Pacific, Middle East, South & Central America, the United States, Canada, other North America,

EU27 (27 members of European Union in 2012), Russia, India, other Europe & Eurasia, and Africa. Appendix A presents the details for spatial aggregation of regions. In Figs. 9(a) and 10(a), the fourteen regions form a circle. The intermediate (or final) export volume of each region is represented by the corresponding arc length around the circle, while the chord that connects two regions represents the intermediate (or final) trade relations between these two linked economies. The color of chord conforms to that of the larger exporter within the two linked regions. In Figs. 9(b) and 10(b), the net intermediate (or final) trade volume of each region is represented by the corresponding arc length around the circle, while the chord that connects two regions represents the net intermediate (or final) trade relations between these two linked economies. The color of the chord conforms to that of the net exporter within the two linked regions.

Fig. 9(a) presents the relations between these fourteen economies in intermediate trade of arable land use. The biggest flow of arable land use in intermediate trade is from Africa to EU27, amounting to 31.66 million ha. As illustrated, while Africa serves the largest exporter in intermediate trade of arable land use, EU27 is illustrated as the biggest receiver. Within Africa's intermediate exports of arable land use (62.61 million ha), around half of them are received by EU27, one-eighth of them going to Japan, one-twelfth to China, one-twelfth to the United States, etc. For the 75.38 million ha of intermediate imports of arable land use by EU27, around 40% of them are from Africa, one-tenth from other Europe & Eurasia, one-tenth from South & Central America, 7% of them from the United States, approximately 5% from China, etc. Meanwhile, among these fourteen economies, EU27 comes as a prominent exporter in intermediate trade of arable land use, second only to Africa. The three major destinations of EU27's intermediate exports of arable land use are China, the United States and other Europe & Eurasia, respectively sharing 15.44%, 16.25% and 21.23% of EU27's intermediate exports.

The net intermediate trade relations between these fourteen economies are presented in Fig. 9(b). Africa is revealed as the biggest

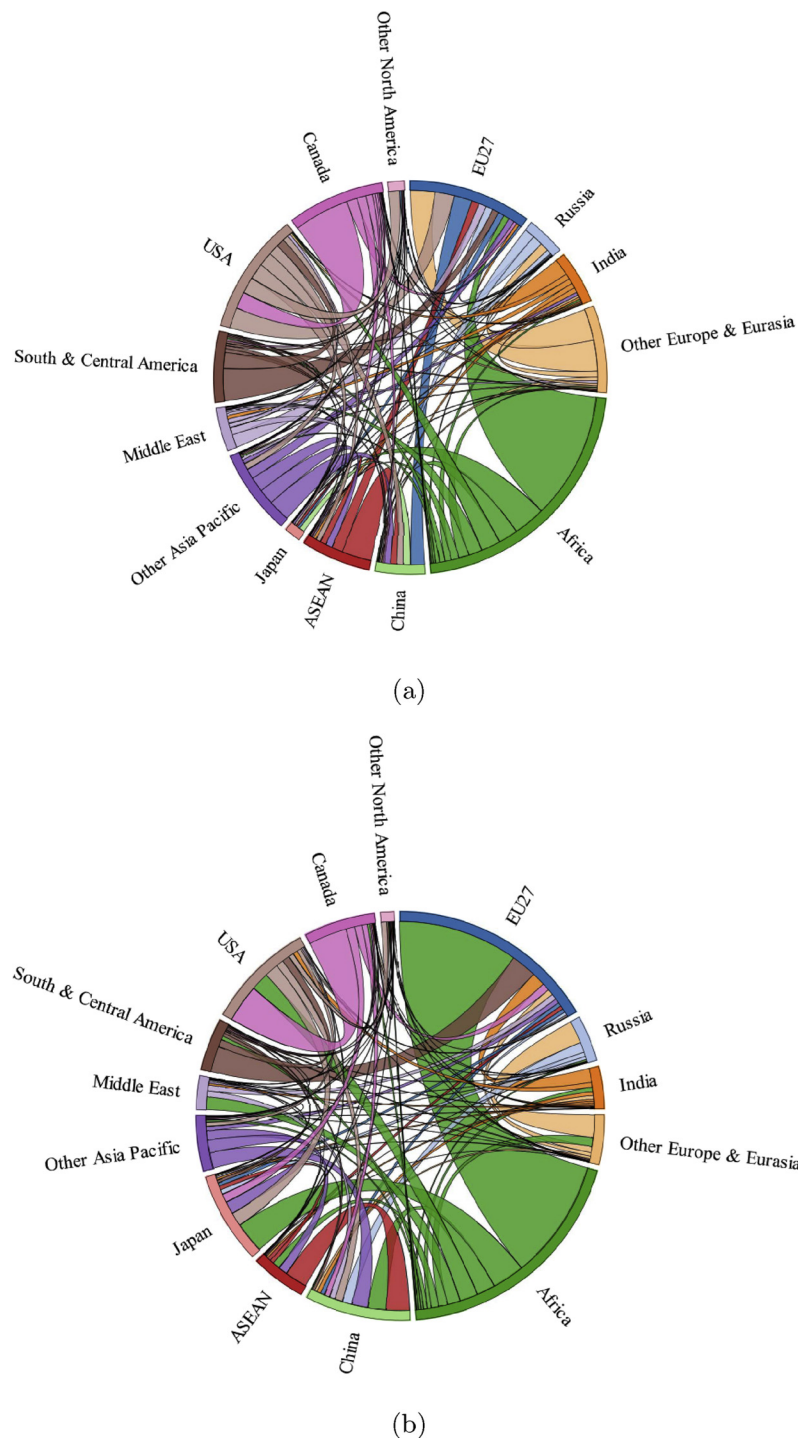


Fig. 9. Intertwined relations of arable land use between the fourteen economies in (a) intermediate trade and (b) net intermediate trade.

deficit region in intermediate trade of arable land use, which obtains an intermediate trade deficit with nearly all of its partners. Inversely, EU27, Japan, and China are shown to have an intermediate trade surplus with all its mates. As demonstrated, the several notable couples of intermediate trade imbalance are Africa-EU27, Canada-the United States, Africa-Japan, South & Central America-EU27, and ASEAN-China. The net intermediate trade flow of arable land use from Africa to EU27 amounts to 30.14 million ha, which is around four times as much as that from Africa to Japan. For the United States, while it maintains a surplus of arable land use in intermediate trade with Canada, Africa, South & Central America and India, it obtains an intermediate trade

deficit with Japan and China. For China, in intermediate trade it mainly receives net inflows of arable land use from ASEAN, other Asia Pacific (mainly Australia), and Africa, respectively amounting to 5.82 million ha, 4.00 million ha and 4.71 million ha.

Fig. 10(a) demonstrates the relations of the fourteen economies in final trade of arable land use, which differ from the intermediate trade connections presented in Fig. 9(a). Africa still maintains the first place in final exports of arable land use (21.23 million ha), with EU27 and India respectively accounting for 51.96% and 12.32% of Africa's arable land outflows. For EU27, its final imports of arable land use from Africa, South & Central America, and India are offset by its final exports to the

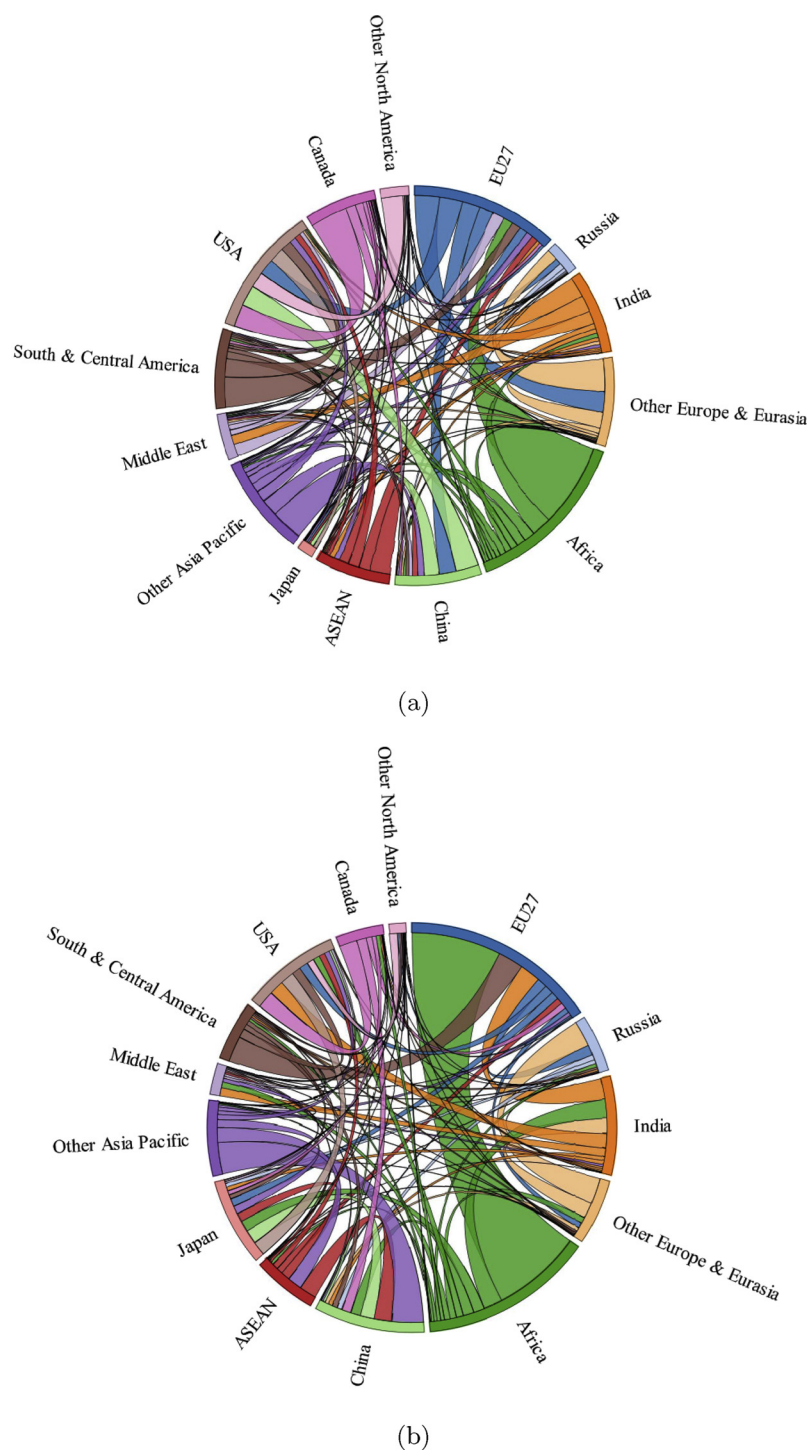


Fig. 10. Intertwined relations of arable land use between the fourteen economies in (a) final trade and (b) net final trade.

United States, China, and Russia. With regard to the United States, it serves dual roles of important importer and exporter of arable land use in final trade. China, Canada, EU27 and South & Central America are illustrated to be important sources of the final imports of arable land use into the United States, and also major destinations of the final exports of arable land use from the United States. For China, it nearly maintains a balance with both the United States and EU27 in final trade of arable land use. For India as another prominent exporter, the arable land use embodied in its final exports goes primarily to EU27, Middle East and the United States.

The net final trade relations between the fourteen economies are

presented in Fig. 10(b). The important couples in terms of final trade imbalance of arable land use are Africa-EU27, China-other Asia Pacific (mainly Australia), EU27-South & Central America, etc. Africa, South & Central America, ASEAN, and Canada remain the major deficit regions while EU27, Japan and China, remain the leading surplus regions in final trade of arable land use.

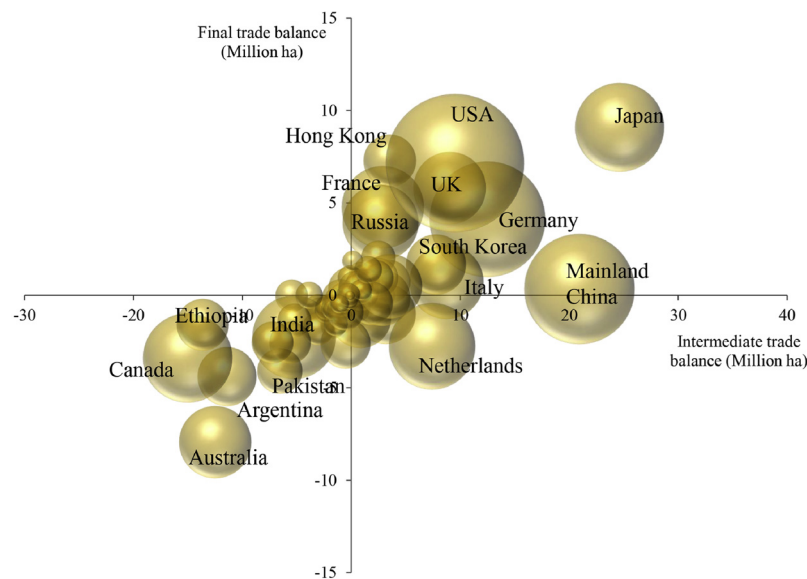


Fig. 11. Trade imbalances of arable land use for regions.

4. Discussion

4.1. Trade imbalance of regions

The trade imbalances of arable land use for regions are discussed in this section. In Fig. 11, the size of the sphere stands for the total trade volume of a region. Intermediate trade balance and final trade balance of arable land use for a region are respectively represented by the net intermediate imports and net final imports of arable land use.

The United States, Germany, Mainland China, Japan, France, Russia, the United Kingdom and Hong Kong, which lie in the first quadrant of the coordinate, are net importers of arable land use in both intermediate trade and final trade. For the United States, though its agricultural population accounts for less than 2% of the national total (BLS, 2016), the United States takes a leading role in agricultural production and food exports on the globe, mainly attributed to the large-scale cultivation & mechanization, as well as the historical series of agricultural policy incentives (such as 1922 Grain Futures Act, 1933 Agricultural Adjustment Act, 2002 Farm Bill, and Food, Conservation and Energy Act of 2008). Renowned for its flourishing agricultural industry, the United States supplies the world market with tremendous agricultural and food products (such as soybeans, corn and wheat) for foreign intermediate production as well as residential consumption, thus exporting a tremendous volume of arable land use abroad. Nevertheless, the exports of arable land use have been offset by the arable land use inflows associated with the massive imported products, including agricultural products, light industry goods (such as toys, furniture, textiles products and plastic bags) and heavy industry products (such as engines, pumps, machines, vehicles, pharmaceuticals and fuel).

For other developed economies such as Germany, the United Kingdom and France, while these regions import lots of agricultural products for further processing in domestic industries, they also require massive products from foreign agricultural sector and food & beverage sector to meet the consumptive demands of local inhabitants, thus gaining a surplus of arable land use in both intermediate and final trades. Meanwhile, at witnessed, apart from the abovementioned developed economies, Mainland China as the largest developing region gains the surplus of arable land use in interregional trade. With only 7% of the world's arable land resources, Mainland China has to feed around 22% of the global population (FAO, 2017). Besides, along with the transformation of a planned economy to a market-based economy, the

rural out-migration not only leads to China's economic miracle in the last three decades, but has caused the massive abandonment & degradation of domestic cultivated land, and aging population of peasants (Gao et al., 2006; Liu, 2014; Liu et al., 2016). Therefore, Mainland China becomes highly reliant on agricultural, and food & beverage products from multiple foreign sources, such as Australia, ASEAN, Africa and Canada, to feed more than one billion population as well as to support domestic intermediate production, thus at the same time bringing home tremendous arable land use.

Inversely, Ethiopia, Tanzania and Ghana, which reside in the third quadrant of the coordinate, are net exporters in both intermediate trade and final trade. Generally, African regions rely heavily on exporting agricultural products to western economies (mainly members of European Union) and eastern regions (such as Mainland China and Japan) and gain a surplus of currency in interregional trade. The price is that tremendous welfare denoted by arable land resources in African regions has been transferred to the comparatively wealthier economies to benefit foreign consumers instead of domestic citizens. Our analysis suggests that the intermediate exports and final exports of arable land use of Africa respectively reach a striking amount of 62.61 million ha and 21.23 million ha. From the long term, this is not a sustainable option for these African regions. Statistics unveil that though Africa is endowed with around one-sixth of global cropland resources, around over half of Africa's arable land is degraded (Cecilie and Anette, 2010). Besides, the rest of the arable land in Africa is exposed to the emerging challenges such as drought and water and soil loss brought about by climate change (Nana-Sinkam, 1995; Yu et al., 2013). Since the limited supply of arable land resource is incapable of feeding the continuous rising population in the Africa regions, millions of people in Africa suffer from famine. In this context, the displacement of local arable land resources for foreign use may exacerbate the tension in satisfying domestic demands in these African regions.

For Canada and Australia as two developed economies, they are also revealed to receive a deficit of arable land use in both intermediate trade and final trade. Due to the endowment of vast soil arable land, these two regions pinpoint agriculture and food & beverage industries as important components of the economy. Serving as a supplier of the agricultural and food products to the United States, Canada transfers massive arable land use to the United States at the same time. While for Australia, the major destinations of its exported agricultural and food & beverage products, such as beans, nuts, oranges, red wine, etc., are mainly Asia Pacific regions such as Mainland China, ASEAN and Japan.

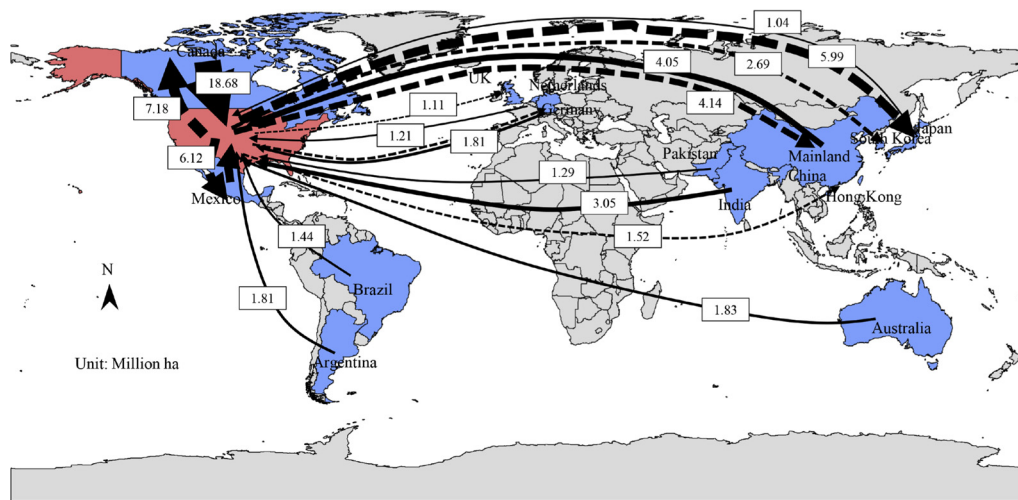


Fig. 12. Geographical details for arable land use embodied in the trade flows of the United States.

Therefore, to some degree, Australia uses its capacious area of arable land to benefit all the consumers in the Asia Pacific regions.

Netherlands as another developed economy situates in the fourth quadrant of the coordinate, which gains a trade surplus in intermediate trade, but a deficit in final trade of arable land use. As a leader in global agricultural and food processing technology, Netherlands is one the several largest exporters of food products. Therefore, on one hand, Netherlands receives large quantities of raw materials and semi-products from foreign agricultural or other sectors for further processing of high value-added food products, thus gaining a surplus of arable land use in intermediate trade. One another hand, Netherlands exports massive high-quality food and agricultural products, such as corn, potatoes, wheat, vegetables, potted plants and flowers, to foreign regions for final consumption, thus becoming a net exporter in final trade.

4.2. Distinct trading economies

In this section, the trade details of the United States and Mainland China as two distinct economies are investigated. Fig. 12 maps the arable land use embodied in the trade flows of the United States. For the arable land use inflows into the United States, those coming from Canada are mostly notable, amounting to 18.68 million ha, followed by those from Mexico (5.60 million ha). Meanwhile, Canada and Mexico as two neighboring nations of the United States, respectively receive 7.18 million ha and 6.12 million ha of arable land use from the United States. India also comes as a main contributor to the arable land use inflows into the United States, with its exports to the United States reaching 3.05 million ha. Meanwhile, while the United States receives 4.05 million ha of arable land use from Mainland China, it also exports 4.14 million ha of arable land use to Mainland China, thus nearly maintaining a bilateral trade balance. As for the United States' important trading partners in Europe, Germany and Netherlands respectively contribute 1.81 million ha and 1.21 million ha to the arable land use imports of the United States. It is also worth noticing that South Korea and Hong Kong remain two important receives of the arable land use outflows from the United States, with the amount reaching 5.99 million ha and 1.52 million ha respectively.

Arable land use inflows and outflows of Mainland China are illustrated in Fig. 13. As witnessed, the United States remains as the biggest market for the arable land use imports of Mainland China. Russia as another leading contributor is revealed to supply 3.35 million ha of arable land use to Mainland China. Other major contributors are Australia, Canada, Ethiopia and Indonesia, whose exports of arable land use to Mainland China are respectively 3.26 million ha, 2.35 million ha, 1.69 million ha and 1.02 million ha. Reputed with a clean environment,

Australia exports massive high-quality products such as wheat, dairy products, infant formula, beef, seafood (such as lobsters, oysters and crabs), wine and fresh fruits (such as table grapes, stone fruits and cherries) to Mainland China, thus at the same time providing Mainland China with the environmental welfare denoted by local arable land resources. For Indonesia, it provides massive land-intensive agricultural products such as animal fats and vegetable oils to Mainland China, thus being a major contributor to the arable land use imports of Mainland China. Ethiopia also serves a big contributor to the arable land inflows into Mainland China by supplying Mainland China with the agricultural and food products such as coffee, sheep and goat meat, dried legumes, and tanned sheep hides. For Germany as the one of Mainland Chinas most important trading partners in Europe, it exports 1.98 million ha of arable land use to Mainland China and also imports 1.37 million ha of arable land use from Mainland China. Meanwhile, it needs to be noted that Mainland China exports 3.45 million ha of arable land use to Japan and receives 1.41 million ha of arable land use from Japan at the same time, thus gaining a deficit of 2.04 million ha of arable land use in bilateral trade with Japan.

4.3. Arable land use self-sufficiency rate by source and that by sink

In this section, two indices in terms of arable land use self-sufficiency rate by source and that by sink are respectively defined, following similar indicators defined for energy use in the study by Chen and Wu (2017). For a region within the world economy, arable land use self-sufficiency rate by source is defined as the ratio of the arable land exploited locally for its own final consumption to the total arable land exploited locally. Arable land use self-sufficiency rate by sink of a region is defined as the arable land exploited locally for its own final consumption to the regions arable land use denoted by arable land use embodied in the products used as its final consumption. The arable land use self-sufficiency rate by source and that by sink for each region are respectively presented in Figs. 14 and 15.

For the United States, its arable land use self-sufficiency rate by source and that by sink are respectively 79.10% and 61.15%. This means that from the supply side, around four fifths of the welfare provided by local arable land resources has finally sank into the products used for domestic final consumption. Besides, the arable land use self-sufficiency rate by sink implies that around two fifths of the United States arable land use stems from the arable land resources exploited from other regions' environmental systems. An interesting picture is observed for Japan, whose arable land use self-sufficiency rate by source and that by sink are respectively 91.77% and 7.93%. As seen, the welfare by domestic arable resources has been nearly all kept at home.

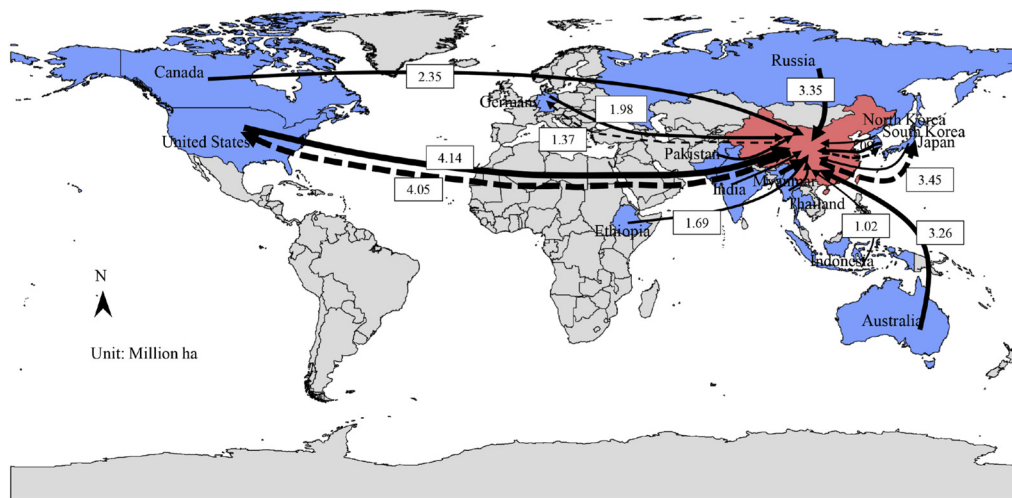


Fig. 13. Geographical details for arable land use embodied in the trade flows of Mainland China.

Besides, over 90% of the arable land use of Japan originates from foreign regions. As seen, as a nation with limited resources, Japan resorts to interregional trade to obtain arable land use from its trading partners to benefit its citizens. For Mainland China and Germany as another two notable importers of arable land use, their arable land use self-sufficiency rates by source are respectively 68.42% and 46.97%, while those by sink are respectively 69.49% and 16.03%. For India as the largest exploiter of arable land resources, its self-sufficiency rate by source and that by sink are respectively 79.92% and 89.56%. For African regions such as Tanzania and Ghana, their arable land use self-sufficiency rates by source are respectively 25.96% and 15.68%, while those by sink are respectively 95.61% and 77.80%. As seen, the arable land use embodied in the final consumption of these African regions mainly comes from local exploitation of arable land. In addition, the majority of local exploitation of arable land resources is embodied in the exported products and finally sinks into foreign regions.

5. Conclusions

In this study, the arable land use for the world economy is tracked from source to sink through interregional trade, by directing equal attention to arable land use embodied in the intermediate inputs as well as that embodied in the primary inputs.

Trade liberalization increases the connections between regions, resulting in the remarkable amount of land displacement worldwide. The total volume of interregional transfer of arable land use amounts to around 40% of the global arable land exploitation. Meanwhile, arable land use embodied in intermediate trade is revealed to be in magnitude

twice that in final trade. Our analysis shows that regions are revealed to behave quite differently in interregional trade of arable land use. Canada, Australia, Argentina, Pakistan and African regions serve as suppliers of arable land use for foreign regions, thus having a massive deficit in both intermediate and final trades. In comparison, the United States, the United Kingdom, Germany and France as notable arable land users displace the arable land use of abovementioned regions via interregional trade in order to sustain the consumptive needs of their local residents, thus obtaining a surplus of arable land use in both intermediate and final trades. Especially, though endowed with sufficient arable land resources, the United States still resorts to interregional trade to obtain the welfare of arable land use from other regions. As revealed in this study, around 40% of the arable land use of the United States originates from land resources in foreign economies.

Meanwhile, it is worth noticing that Mainland China as the largest developing economy is also revealed as a net importer of arable land use in intermediate trade as well as in final trade. Though Mainland China has a long history of agricultural foundations, the local arable land resources have been severely challenged by the outflow of peasants and evasion of built-up areas with the ongoing urbanization. Therefore, Mainland China has become a major importer of agricultural products in the world, thus in an indirect way bringing in the welfare endowed by arable land resources from other regions. Besides, apart from the agricultural products, Mainland China as the “world factory” imports vast land-intensive materials, semi-products and equipment from abroad for further domestic reprocessing, thus making it the second biggest net importer of arable land use in intermediate trade. Our analysis suggests that African regions (such as Tanzania and

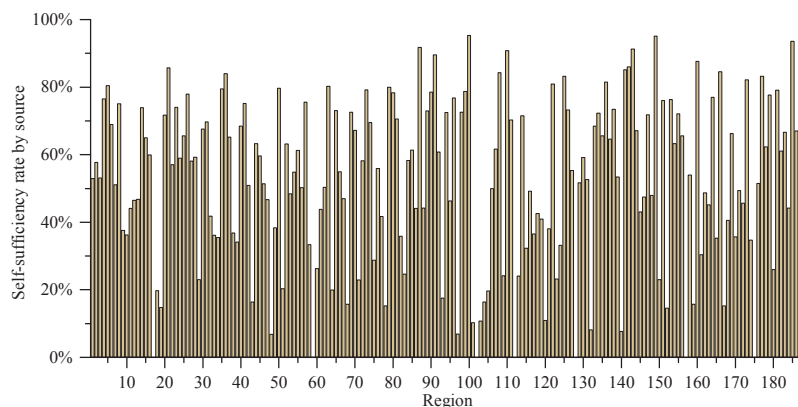


Fig. 14. Arable land use self-sufficiency rate by source for each region.

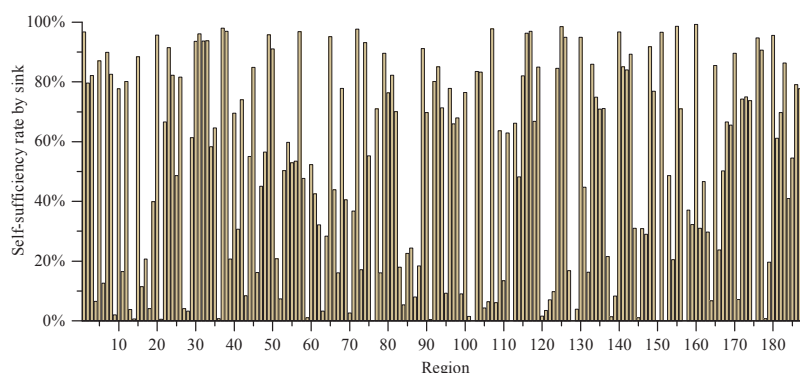


Fig. 15. Arable land use self-sufficiency rate by sink for each region.

Ethiopia), Indonesia, Canada, Australia and Russia are the main contributors to the arable land use inflows into Mainland China. Meanwhile, it is also noticing that the per-capita arable land use of Mainland China is below the world average and much lower in contrast to that of the developed economies, which leaves sufficient room for potential increase. Today the burgeoning middle class in Mainland China is becoming wealthy and aspiring for enhanced living standards. They have a preference for high-quality and nutritious food that is imported from abroad. It could be foreseen that interregional trade will gradually become a major tool for Mainland China to displace the use of arable land resources in foreign regions.

Overall, the diminishing arable land resources have become a global concern. The period is gone when regions make use of foreign arable land resources by establishing overseas colonies. In the globalized world economy, the invisible cross-border transfer of arable land use associated with traded commodities will become the primary means for a region to acquire the environmental welfare from other regions. In this context, the regions shall put themselves in the global context and adjust both intermediate and final trade patterns to be more adapted to the global supply chain as well as to benefit from interregional trade. Eyes will not only be focused on the economic trade balances, but also on the trade balance of arable land use that matters greatly to the sustainability of resource use.

Conflict of interest

None.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.landusepol.2018.05.005>.

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